A Frequency Domain Method for Blind Source Separation of Convolutive Audio Mixtures

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***Abstract*—In this paper, we propose a new frequency domain approach to blind source separation (BSS) of audio signals mixed in a reverberant environment. We propose a joint diagonalization procedure on the cross power spectral density matrices of the signals at the output of the mixing system to identify the mixing system at each frequency bin up to a scale and permutation ambiguity. The frequency domain joint diagonalization is performed using a new and quickly converging algorithm which uses an alternating least-squares (ALS) optimization method. The inverse of the mixing system is then used to separate the sources. An efficient dyadic algorithm to resolve the frequency dependent permutation ambiguities that exploits the inherent nonstationarity of the sources is presented. The effect of the unknown scaling ambiguities is partially resolved using an initialization procedure for the ALS algorithm. The performance of the proposed algorithm is demonstrated by experiments conducted in real reverberant rooms. Performance comparisons are made with previous methods.**

***Index Terms*—Audio enhancement, frequency domain blind source separation, joint diagonalization, permutation ambiguity.**

I. Introduction

In the blind source separation (BSS) problem, the objective is to separate multiple sources, mixed through an unknown mixing system (channel), using only the system output data (observed signals) and in particular without using any (or the least amount of) information about the sources or the system. The blind source separation problem arises in many fields of study, including speech processing, data communication, biomedical signal processing, etc. A difficult practical example for the convolutive BSS problem is separation of audio signals mixed in a reverberant environment.

In recent years, a few BSS methods have been proposed that exploit nonstationarity of the source signals. A nonstationarity assumption can be justified by realizing that most real world signals are inherently nonstationary (e.g., speech or biological signals). In [1]–[3], methods have been proposed for BSS of instantaneously mixed, nonstationary signals (cyclostationary in the second reference), while [4] considers BSS of colored nonstationary signals for the convolutive case. A Bayesian approach based on nonstationarity of the source is considered in [5].

Approaches for the convolutive case can be divided into frequency domain [4], [6], [7] and time domain methods [8]–[10]. The advantage of using frequency-domain methods is that a time-domain problem with a large number of parameters is decomposed into multiple, independent estimation problems at each frequency bin, each with fewer parameters to be estimated. As a result, in general the frequency-domain estimation algorithms have a simpler implementation and better convergence properties over their time-domain counterparts. The main difficulties with frequency-domain BSS of convolutive mixtures, however, are the arbitrary permutation and scaling ambiguities of the estimated frequency response of the un-mixing system at each frequency bin.

The method proposed in this paper is a frequency domain technique that exploits second-order nonstationarity of the input signals to separate convolved audio sources, for the case where the mixing system is stationary. Other methods, such as [4], [11], also use a similar approach. The method offered in this paper differs from previous works in at least three respects. First, the previous approaches are based on a gradient descent procedure, while the method proposed in this paper is based on a more efficient alternating least-squares with projection (ALSP) method. A proof of convergence of this algorithm to at least a local minimum is provided. Second, an efficient method for addressing the permutation problem inherent with frequency domain approaches is proposed. This algorithm exploits the spectral correlation of the sources between adjacent frequency bins. Most audio signals of interest possess these properties. Other researchers, e.g., [11] and [12], have proposed similar ideas for the permutation problem. The difficulty is that an error in the permutation correction scheme of these previous approaches can have a catastrophic impact on the quality of the output signals. However, the technique presented in this paper modifies past approaches so that the impact of such errors can be reduced. Thirdly, the scaling problem, also inherent with frequency domain approaches, is partially addressed in this paper by an initialization procedure proposed previously; e.g., [12], [13]. However, in the former reference, the initialization procedure was used to reduce permutation error, not to address the scaling issue as in our case. The combination of the approaches adopted in this paper significantly improves the quality of the separated audio, as has been demonstrated in our real room audio experiments.

Further, in this paper we show our results for real room experiments with typical reverberation times. To date, there have been relatively few methods that consider the real room situation; examples are [4], [11], [14]. The proposed method is not limited to mixing systems with fixed dimensions nor to ones with the same number of outputs and inputs. The only requirement is that the number of outputs for the mixing system be greater than or equal to the number of inputs.

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A frequency domain blind identification method also exploiting nonstationarity has been previously discussed in [15], [16]. These works require that the mixing system be column-wise co-prime and the sources white, in order to identify the mixing system up to a frequency *independent* permutation and scale ambiguity. However, in the audio context when dealing with reverberant enclosures, these conditions do not necessarily hold. In this paper, these previous works are extended to apply to the audio case.

The organization of this paper is as follows: The problem formulation including the set of required assumptions is presented in Section II. Section III discuses the joint diagonalization problem. In Section IV we present a frequency-domain algorithm for convolutive BSS. A novel solution to the permutation problem is discussed in Section V. Simulation results for synthetic convolutive mixing scenarios are described in Section VI and real room experimental results are described in Section VII. The first set of experiments are based on real room recordings done in a small office environment with a moderate reverberation time. The second set of experiments is done in a more acoustically live conference room with a larger reverberation time. Additional results based on real-room recordings are also included to demonstrate the robustness of the proposed permutation algorithm. The performance of the proposed method is compared with that of [4]. Conclusions and final remarks are presented in Section VIII.

II. Problem Statement

A. Notation

We use bold upper and lower case letters to show matrices and vectors respectively. The remaining notational conventions and major symbols are listed as follows.

$(\cdot)^*$	Complex Conjugation.
$(\cdot)^T$	Transpose.
$(\cdot)^\dagger$	Hermitian transpose.
$E\{\cdot\}$	Expectation Operator.
diag$\{\mathbf{a}\}$	Forms a diagonal matrix from the vector $\mathbf{a}$.
diag$\{\mathbf{A}\}$	Forms a column vector from the diagonal elements of $\mathbf{A}$.
vec$\{\mathbf{A}\}$	Forms a column vector by stacking the columns of $\mathbf{A}$.
mat$\{\mathbf{a}\}$	Forms a $J \times J$ matrix from a $J^2 \times 1$ column vector $\mathbf{a}$.
$\mathbf{A}^+$	Pseudo inverse of the matrix $\mathbf{A}$.
J	Number of sensors.
K	Number of frequency bins.
L	Length of the mixing system.
M	Number of epochs.
N	Number of sources.

B. Convolative Mixing Problem

We consider the following N-source J-sensor multi-input multi-output (MIMO) linear model for the received signal for the convolutive mixing problem:[1]

$$\mathbf{x}(t) = [\mathbf{H}(t)] * \mathbf{s}(t) + \mathbf{n}(t) \quad t \in \mathbb{Z} \tag{1}$$

where $\mathbf{x}(t) = (x_1(t), \cdots, x_J(t))^T$ is the vector of observed signals, $\mathbf{s}(t) = (s_1(t), \cdots, s_N(t))^T$ is the vector of sources, $\mathbf{H}(t)$ is the $J \times N$ impulse response of the mixing system, and $\mathbf{n}(t) = (n_1(t), \cdots, n_J(t))^T$ is the additive noise vector. We assume $h_{ij}(z)$, the ij_{th} element of $\mathbf{H}(z)$ which is the z-transform of $\mathbf{H}(t)$, to be a rational function of z.

The objective of BSS may be stated in the frequency domain, as estimating the un-mixing filters $\mathbf{W}(\omega)$ from the observed signals $\mathbf{x}(t)$ such that

$$\mathbf{W}(\omega)\mathbf{H}(\omega) = \mathbf{\Pi}\mathbf{D}(\omega) \quad \forall \omega \in [0, \pi) \tag{2}$$

where $\mathbf{\Pi} \in \mathbb{R}^{N \times N}$ is a permutation matrix, $\mathbf{D}(\omega) \in \mathbb{C}^{N \times N}$ is a diagonal matrix with diagonal elements which are rational functions of the radian frequency ω, and $\mathbf{H}(\omega) = \mathbf{H}(z)|_{z=e^{j\omega}}$. Equation (2) represents a composite mixing-unmixing system which recovers the sources up to frequency-independent permutation but frequency-dependent scale ambiguities.

Typically with frequency-domain methods, the matrix $\mathbf{\Pi}$ is dependent on ω, thereby introducing frequency-dependent permutation errors in the output frequency response. Therefore, to ensure adequate separation performance, $\mathbf{\Pi}$ must be made independent of frequency (as it is in (2)). The presence of the frequency-dependent quantity $\boldsymbol{D}(\omega)$ in (2) is a result of an inherent scale or phase ambiguity at each frequency ω. Thus, the outputs of the un-mixing filter, although separated, are a filtered version of the original sources. To ensure satisfactory audio output quality, $\boldsymbol{D}(\omega)$ should at least be approximately constant with frequency.

C. Main Assumptions

- A0: $J \geq N \geq 2$; i.e, we have at least as many sensors as sources and number of the sources are at least two.
- A1: The sources $\mathbf{s}(t)$ are zero mean, second-order quasistationary signals[2] and the cross-spectral density matrices of the sources $\mathbf{P}_s(\omega, m)$ are diagonal for all ω and m where m is the epoch[3] index.
- A2: The mixing system is modeled by a causal system of the form $\mathbf{H}(z) = [\mathbf{h}_1(z), \ldots, \mathbf{h}_N(z)]$. The elements of $\mathbf{H}(z)$ are rational functions of z, and $\mathbf{H}$ does not change over the entire observation interval.
- A3: $\mathbf{H}(\omega_k)$, the discrete Fourier transform (DFT) of $\mathbf{H}(t)$, has full column rank for all ω_k, $k = 0, \ldots, K-1$, $\omega_k = 2\pi k/K$. Also $||\mathbf{h}_i(\omega_k)||_2^2 = 1$, where $\mathbf{h}_i(\omega_k)$ is the i_{th} column of $\mathbf{H}(\omega_k)$.
- A4: The noise $\mathbf{n}(t)$ is zero mean, i.i.d. across sensors, with power σ^2. The noise is assumed independent of the sources.

A1 is the core assumption. As is shown in [4] and [16], this nonstationarity assumption enables us to separate the convolved sources by applying joint diagonalization on the second-order statistics of the observed signal, and makes it possible to resolve the frequency-dependent permutations. The first part of

[1]Here, we use the notation $[\mathbf{H}(t)] * \mathbf{s}(t)$ to denote the convolution between a matrix system with impulse response $\mathbf{H}(t)$ and the source vector $\mathbf{s}(t)$

[2]By second order quasistationarity we mean the variances of the signals are slowly varying with time such that over some short time interval (an "epoch"), they can be assumed approximately stationary.

[3]By "epoch", we mean a duration of time for which the source signals may be considered approximately stationary within the epoch, but nonstationary between different epochs.

assumption *A3* guarantees that $\mathbf{H}(\omega_k)$ is invertible for all $k = 0,\ldots,K-1$. Although there is no physical justification for the second part of *A3*, we use it to resolve the inherent scaling ambiguities that exist in our algorithm to identify $\boldsymbol{H}(\omega)$.

Let $\tilde{\mathbf{P}}_x(\omega,m)$ represent the true cross-spectral density matrix of the observed signal at frequency ω and time epoch m. Based on the above assumptions, we have

$$\tilde{\mathbf{P}}_x(\omega,m) = \mathbf{H}(\omega)\mathbf{P}_s(\omega,m)\mathbf{H}^\dagger(\omega) + \sigma^2\mathbf{I}. \tag{3}$$

For $J > N$, σ^2 is given as the smallest eigenvalue of the matrix $\tilde{\mathbf{P}}_x(\omega,m)$. Therefore a noise-free cross-spectral density matrix can be obtained as follows:

$$\mathbf{P}_x(\omega,m) = \tilde{\mathbf{P}}_x(\omega,m) - \sigma^2\mathbf{I} = \mathbf{H}(\omega)\mathbf{P}_s(\omega,m)\mathbf{H}^\dagger(\omega). \tag{4}$$

In practice, we use the discrete frequency variable $\omega_k = 2\pi k/K$ instead of the continuous variable ω to calculate an estimate $\hat{\mathbf{P}}(\omega_k,m)$ of $\mathbf{P}_x(\omega,m)$. The estimation of $\mathbf{P}_x(\omega,m)$ is discussed in the Appendix.

III. Joint Diagonalization Problem

The first stage of the proposed algorithm employs joint diagonalization of the set of estimated cross power spectral density matrices $\hat{\mathbf{P}}_x(\omega_k,m)$, $m = 0,\ldots,M-1$ at each frequency ω_k, over M epochs, to estimate the mixing system up to a permutation and diagonal scaling ambiguity at each frequency bin.

The joint diagonalization problem was first introduced by Flurry [17] and later on was used as a tool for solving the BSS problem by [4], [18]–[22]. The problem is expressed as finding a single matrix $\mathbf{Z}$ that jointly (approximately) diagonalizes the set of matrices $\mathbf{R}_1,\ldots,\mathbf{R}_M$. The most common criterion used for joint diagonalization is the one given as

$$\arg\min_{\mathbf{Z}} \sum_{m=1}^{M} \mathrm{Off}(\mathbf{Z}^\dagger\mathbf{R}_m\mathbf{Z}) \tag{5}$$

where $\mathrm{Off}(\mathbf{X})$ for an arbitrary matrix $\mathbf{X}$ is defined as the sum squares of the off-diagonal values of the matrix $\mathbf{X}$. Another common criterion is the following least-squares cost function

$$\arg\min_{\mathbf{A},\boldsymbol{\Lambda}(m)} \sum_{m=1}^{M} \left\|\mathbf{R}_m - \mathbf{A}\boldsymbol{\Lambda}(m)\mathbf{A}^\dagger\right\|_F^2 \tag{6}$$

where $\boldsymbol{\Lambda}(m)$ is diagonal for all m, and $\|\cdot\|_F$ denotes the Frobenius norm.

By using a joint diagonalization procedure, the mixing system $\mathbf{H}(\omega_k)$ can be estimated up to a frequency dependent permutation $\boldsymbol{\Pi}(\omega_k)$ and frequency dependent scale ambiguity $\mathbf{D}(\omega_k)$. That is, at each frequency ω_k we substitute the set of matrices $\mathbf{R}_1,\ldots,\mathbf{R}_M$ in (6) with the set $\mathbf{P}_x(\omega_k,m)$, $m = 0,\ldots,M-1$, defined by (4). (Note that this sequence of matrices is a consequence of the nonstationarity of the sources). Then, if we find a matrix $\mathbf{B}(\omega_k)$ and diagonal matrices $\boldsymbol{\Lambda}(\omega_k,m)$ such that $\mathbf{P}_x(\omega_k,m) = \mathbf{B}(\omega_k)\boldsymbol{\Lambda}(\omega_k,m)\mathbf{B}^\dagger(\omega_k)$ with the scale constraint $\|\mathbf{b}_i(\omega_k)\|_2^2 = 1$, where $\mathbf{b}_i(\omega_k)$ is the i_{th} column of $\mathbf{B}(\omega_k)$, then the following relation holds:

$$\mathbf{B}(\omega_k) = \mathbf{H}(\omega_k)\boldsymbol{\Pi}(\omega_k)\mathbf{D}(\omega_k) \tag{7}$$

where the diagonal entries of $\mathbf{D}(\omega_k)$ are of the form $e^{j\gamma_k}$, where γ_k is a phase [15]. A procedure to resolve the frequency dependent permutation ambiguity is given in Section V. A partial solution to the frequency-dependent phase ambiguity problem is discussed in Section IV-A.

We now discuss the procedure for determining the unmixing system $\mathbf{W}(t)$ given the $\mathbf{B}(\omega_k)$, $k = 0,\ldots,K-1$. This procedure has been previously used, e.g., in [23]. For this presentation, we assume the permutations and scale ambiguities are correctly resolved. At each frequency, we calculate $\mathbf{W}(\omega_k)$ from

$$\mathbf{W}(\omega_k) = \mathbf{B}^+(\omega_k) \tag{8}$$

where $\mathbf{B}^+(\omega_k)$ is the pseudo inverse of the matrix $\mathbf{B}(\omega_k)$. For $J \geq N$, $\mathbf{W}(\omega_k)\mathbf{B}(\omega_k) = \mathbf{I}_N$ [24], where $\mathbf{I}_N$ is the $N \times N$ identity matrix. Since the composite multi-dimensional mixing-unmixing system is an identity at each frequency, the sources are recovered at the outputs at each frequency. The unmixing system $\mathbf{W}(t)$ is then formed as the inverse DFT of $\mathbf{W}(\omega_k)$, $k = 0,\ldots,K-1$. In general, the $\mathbf{W}(t)$ is neither causal nor of finite length.[4] $\mathbf{W}(t)$ may be made causal by imposing a suitable delay. The effect of time-domain aliasing error in the inverse DFT, induced by the infinite length of $\mathbf{W}(t)$, can be made negligible by choosing K, which is the number of frequency bins and also the length of $\mathbf{W}(t)$, to be much greater than L, the length of $\mathbf{B}(t)$.

IV. Algorithm

Based on the joint diagonalization principle, we propose the following least-squares joint diagonalization criterion by analogy to (6), for the case when a sample estimate $\hat{\mathbf{P}}_x(\omega_k,m)$ of each $\mathbf{P}_x(\omega_k,m)$ is available:

$$\min_{\substack{\mathbf{B}(\omega_k),\boldsymbol{\Lambda}(m)\\ \|\mathbf{b}_i(\omega_k)\|_2=1}} \sum_{k=0}^{K-1}\sum_{m=0}^{M-1} \left\|\hat{\mathbf{P}}_x(\omega_k,m) - \mathbf{B}(\omega_k)\boldsymbol{\Lambda}(\omega_k,m)\mathbf{B}^\dagger(\omega_k)\right\|_F^2 \tag{9}$$

where $\boldsymbol{\Lambda}(\omega_k,m)$ is a diagonal matrix representing the unknown cross-spectral density matrix of the sources at epoch m.

In [4], a similar criterion has been proposed that directly estimates the separating matrix $\mathbf{W}(\omega_k)$. Using the criterion in (9) allows us to implement the ALS algorithm as will be described later in this section. In [4], an additional FIR constraint on the length of the un-mixing matrix is required to prevent arbitrary frequency dependent permutations. However, as shown in [25] and [26], such a constraint is not effective for long reverberant environments and the performance of the algorithm may degrade as the length of the separating filter increases. In the proposed method we do not require an FIR length constraint on the mixing model, mainly because we use a different dyadic approach for resolving the permutation problem, that exploits the inherent nonstationarity of the sources.

For the first stage of the algorithm we optimize the criterion given by (9) using an alternating least-squares (ALS) approach. The basic idea behind the ALS algorithm is that in the optimization process we divide the parameter space into multiple

[4]However, since the energy in $\mathbf{W}(t)$ must be bounded, this signal must decay toward zero as time increases.

sets. At each iteration of the algorithm we minimize the criterion with respect to one set conditioned on the previously estimated sets of parameters. The newly estimated set is then used to update the remaining sets. This process continues until convergence is achieved. Alternating least-squares methods have been used for BSS of finite alphabet signals in [27] and [28] and *parallel factor analysis* (PARAFAC) in [29]. The advantage of using ALS (rather than gradient based optimization methods) is that it usually has fast convergence (as will be demonstrated in the simulations) and there are no parameters to adjust. One disadvantage, which is also shared by gradient techniques for nonlinear optimization, is that unless it is properly initialized, it can fall into a local minimum. Later in this section we introduce a procedure for initializing the algorithm to diminish this possibility.

Referring back to criterion (9), and using the properties of Kronecker products [30], the quantity $\mathbf{B}(\omega_k)\mathbf{\Lambda}(\omega_k,m)\mathbf{B}^\dagger(\omega_k)$ in (9) can be written as

$$\text{vec}\left\{\mathbf{B}(\omega_k)\mathbf{\Lambda}(\omega_k,m)\mathbf{B}^\dagger(\omega_k)\right\} = \left[\mathbf{B}(\omega_k)\odot\mathbf{B}^\dagger(\omega_k)\right] \times\text{diag}\left\{\mathbf{\Lambda}(\omega_k,m)\right\} \quad (10)$$

where $\odot$ is the Khatri–Rao product and is defined as

$$\mathbf{B}(\omega_k)\odot\mathbf{B}^\dagger(\omega_k)=[\mathbf{b}_1(\omega_k)\otimes\mathbf{b}_1^*(\omega_k),\ldots,\mathbf{b}_N(\omega_k)\otimes\mathbf{b}_N^*(\omega_k)] \quad (11)$$

where $\otimes$ represents the Kronecker product. Setting $\mathbf{G}(\omega_k)=\mathbf{B}(\omega_k)\odot\mathbf{B}(\omega_k)$, $\mathbf{d}(\omega_k,m)=\text{diag}\{\mathbf{\Lambda}(\omega_k,m))\}$ and $\hat{\mathbf{p}}_x(\omega_k,m)=\text{vec}\{\hat{\mathbf{P}}_x(\omega_k,m)\}$ we can rewrite (9) as:

$$\arg\min_{\mathbf{g}_i(\omega_k)\in\Omega,\mathbf{d}(\omega_k,m)}\sum_{k=0}^{K-1}\sum_{m=0}^{M-1}\|\hat{\mathbf{p}}_x(\omega_k,m) -\mathbf{G}(\omega_k)\mathbf{d}(\omega_k,m)\|_2^2,\quad i=1,\ldots,N \quad (12)$$

where $\mathbf{g}_i(\omega_k)$ is the i_{th} column of $\mathbf{G}(\omega_k)$. Based on assumption *A3* we have $\|\mathbf{b}_i(\omega_k)\|_2^2=1$ and we define the constraint set $\Omega\subset\mathbb{C}^{J^2\times 1}$ as

$$\Omega=\left\{\text{vec}\{\mathbf{\Phi}\}|\mathbf{\Phi}=\boldsymbol{\nu}\boldsymbol{\nu}^\dagger,\ \boldsymbol{\nu}\in\mathbb{C}^{J\times 1},\ \|\boldsymbol{\nu}\|_2^2=1\right\}. \quad (13)$$

In defining the constraint set Ω, we used the fact that for a column vector $\boldsymbol{\nu}$, we have $\boldsymbol{\nu}\otimes\boldsymbol{\nu}=\text{vec}\{\boldsymbol{\nu}\boldsymbol{\nu}^\dagger\}$. Following the discussion above, we can first minimize (12) with respect to $\mathbf{g}_i(\omega_k)$ conditioned on $\hat{\mathbf{d}}(\omega_k,m)$ and $\hat{\mathbf{g}}_j(\omega_k)$, respectively. These latter quantities are the previously estimated values of $\mathbf{d}(\omega_k,m)$ and $\mathbf{g}_j(\omega_k)$, $j\neq i$, $j=1,\cdots,N$. To do this, we first form the matrix

$$\mathbf{F}_i(\omega_k)=\mathbf{T}(\omega_k)-\sum_{\substack{j=1\\ j\neq i}}^{N}\hat{\mathbf{g}}_j(\omega_k)\mathbf{z}_j^\dagger(\omega_k) \quad (14)$$

where $\mathbf{z}_j(\omega_k)=(\hat{d}_j(\omega_k,0),\ldots,\hat{d}_j(\omega_k,M-1))^T$ and $\mathbf{T}(\omega_k)=[\hat{\mathbf{p}}(\omega_k,0),\ldots,\hat{\mathbf{p}}(\omega_k,M-1)]$.

According to the ALS procedure, we now address the optimization of (12) with respect to the $\mathbf{g}_i$. Using (14), this optimization is equivalent to

$$\min_{\mathbf{g}_i(\omega_k)\in\Omega}\sum_{k=0}^{K-1}\left\|\mathbf{F}_i(\omega_k)-\mathbf{g}_i(\omega_k)\mathbf{z}_i^\dagger(\omega_k)\right\|_F^2. \quad (15)$$

Notice that (15) is a constrained least-squares problem. To solve this problem we use the following lemma proved in [31].

Lemma 1: The constrained least squares problem $\min_{\mathbf{x}\in\Omega}\|\mathbf{A}-\mathbf{x}\mathbf{b}^\dagger\|_F^2$ where $\mathbf{A}\in\mathbb{C}^{M\times N}$, $\mathbf{x}\in\mathbb{C}^{M\times 1}$, $\mathbf{b}\in\mathbb{C}^{N\times 1}$ and Ω is the constraint set, is equivalent to

$$\min_{\mathbf{x}\in\Omega}\|\mathbf{x}-\tilde{\mathbf{x}}\|_2^2 \quad (16)$$

where $\tilde{\mathbf{x}}$ is the unconstrained solution to $\|\mathbf{A}-\mathbf{x}\mathbf{b}^\dagger\|_F^2$, given by

$$\tilde{\mathbf{x}}=\frac{\mathbf{A}\mathbf{b}}{\mathbf{b}^\dagger\mathbf{b}}. \quad (17)$$

■

Based on the above lemma, to minimize (15) we first find the unconstrained least-squares minimizer of (15) by setting

$$\tilde{\mathbf{g}}_i(\omega_k)=\frac{\mathbf{F}_i(\omega_k)\mathbf{z}_i(\omega_k)}{\mathbf{z}_i^\dagger(\omega_k)\mathbf{z}_i(\omega_k)}. \quad (18)$$

We then project $\tilde{\mathbf{g}}_i(\omega_k)$ onto Ω through following minimization:

$$\hat{\mathbf{g}}_i(\omega_k)=\arg\min_{\mathbf{g}_i(\omega_k)\in\Omega}\|\tilde{\mathbf{g}}_i(\omega_k)-\mathbf{g}_i(\omega_k)\|_2^2. \quad (19)$$

Since $\mathbf{g}_i(\omega_k)=\text{vec}\{\mathbf{b}_i(\omega_k)\mathbf{b}_i^\dagger(\omega_k)\}$, by defining $\mathbf{Y}_i(\omega_k)=\text{mat}\{\tilde{\mathbf{g}}_i(\omega_k)\}$, we can write the above equation as

$$\begin{aligned}&\min_{\|\mathbf{b}_i(\omega_k)\|_2=1}\left\|\mathbf{Y}_i(\omega_k)-\mathbf{b}_i(\omega_k)\mathbf{b}_i^\dagger(\omega_k)\right\|_F^2\\ &=\min_{\|\mathbf{b}_i(\omega_k)\|_2=1}\left(\mathbf{b}_i^\dagger(\omega_k)\mathbf{b}_i(\omega_k)\right)^2\\ &\quad+\text{Trace}\left\{\mathbf{Y}_i^\dagger(\omega_k)\mathbf{Y}_i(\omega_k)\right\}-2\mathbf{b}_i^\dagger(\omega_k)\mathbf{Y}_i(\omega_k)\mathbf{b}_i(\omega_k)\\ &=\min_{\|\mathbf{b}_i(\omega_k)\|_2=1}C-2\mathbf{b}_i^\dagger(\omega_k)\mathbf{Y}_i(\omega_k)\mathbf{b}_i(\omega_k)\end{aligned} \quad (20)$$

where $C=1+\text{Trace}\{\mathbf{Y}_i^\dagger(\omega_k)\mathbf{Y}_i(\omega_k)\}$ is a constant term. The above minimization can be done easily by choosing $\hat{\mathbf{b}}_i(\omega_k)$, the estimated i_{th} column of $\mathbf{B}(\omega_k)$, to be the dominant eigenvector of $\mathbf{Y}_i(\omega_k)$. To find the dominant eigenvector of a matrix we can use the power iteration method described in [24]. Since an initial estimate of $\mathbf{b}_i(\omega_k)$ is available (as is explained later), $\mathbf{Y}_i(\omega_k)$ is nearly a rank one matrix. Hence, the ratio of the largest eigenvalue of $\mathbf{Y}_i(\omega_k)$ to the second-largest (this ratio determines the convergence of the power method), is large. Hence, we need to apply only few iterations of the power method to minimize (20).[5]

After all the $\mathbf{g}_i(\omega_k)$, $i=1,\ldots,N$ have been estimated, based on the method described above, we then minimize (9) with respect to $\mathbf{d}(\omega_k,m)$ conditioned on the previous estimate of $\mathbf{G}(\omega_k)$. We solve the following least-squares problem:

$$\hat{\mathbf{d}}(\omega_k,m)=\arg\min_{\mathbf{d}(\omega_k,m)}\left\|\hat{\mathbf{p}}_x(\omega_k,m)-\hat{\mathbf{G}}(\omega_k)\mathbf{d}(\omega_k,m)\right\|_2^2. \quad (21)$$

Minimizing (21) with respect to $\mathbf{d}(\omega_k,m)$, we get

$$\hat{\mathbf{d}}(\omega_k,m)=\hat{\mathbf{G}}^+(\omega_k)\hat{\mathbf{p}}(\omega_k,m) \qquad m=0,\ldots,M-1,\ k=0,\ldots,K-1. \quad (22)$$

[5]In our simulations we use only one power iteration per ALS iteration. Increasing the number of iterations beyond one did not noticeably improve the convergence nor the performance of the algorithm.

Using (18), (19), and (22), we can repeatedly update the values of $\mathbf{d}(m)$ and $\mathbf{g}_i(\omega_k)$, $i = 1, \ldots, N$ until convergence is achieved.

Convergence: The convergence of above algorithm follows from Lemma 1 by noticing that each step of the algorithm either decreases or maintains the value of cost function. Since the cost function is bounded from below, convergence results. ■

As mentioned previously, to avoid being trapped in local minima, and to ensure perceptually acceptable voice quality, we need to properly initialize the ALS algorithm. As outlined in the following discussion, we note that the initialization procedure has a significant impact on resolving the scaling ambiguity problem.

A. Initialization

To initialize the algorithm we have different options. The first option is that a rough estimate of the mixing system at each frequency bin (up to some scaling and permutation ambiguity) can be obtained using the following closed-form, exact joint diagonalization procedure [32].

Select two matrices $\hat{\mathbf{P}}(\omega_k, m_1)$, $\hat{\mathbf{P}}(\omega_k, m_2)$, $m_1 \neq m_2$. Then choose the initial estimate for $\mathbf{B}(\omega_k)$ to be the matrix consisting of the generalized eigenvectors of the matrix $\hat{\mathbf{P}}(\omega_k, m_2)\hat{\mathbf{P}}^{-1}(\omega_k, m_1)$. For the case where $\mathbf{B}(\omega_k)$ is not square, the method proposed in [33] can be used. Although no optimum selection for m_1 and m_2 can be given at this stage, in our simulations we choose $\hat{\mathbf{P}}(\omega_k, m_1)$ and $\hat{\mathbf{P}}(\omega_k, m_2)$ such that their nonzero generalized eigenvalues are not all repeated. Since we need to use this initialization procedure at each frequency bin, one draw-back of above initialization method will be its computational complexity.

An alternative, *ad hoc* initialization method which not only requires less computation, but also improves the quality of the separated audio signals, is described as follows. This method is similar to that proposed in [12]. The main idea of this initialization procedure, is that first we choose the initial value of $\mathbf{B}(\omega_0)$, the first frequency bin, using the exact closed form joint diagonalization method described above. We then apply the ALS algorithm to find the final estimate of $\mathbf{B}(\omega_0)$. This final estimate is then used as an initial value for the next adjacent frequency bin, which is $\mathbf{B}(\omega_1)$. The outcome of this frequency bin is also used as an initial value for the next frequency bin and this procedure continues until all the frequency bins have been covered. Note that in this way we need to apply the exact closed form joint diagonalization algorithm only for one frequency bin.

As is demonstrated in our simulation results, this initialization procedure results in improved quality of the separated audio signals. An intuitive explanation for this is as follows. We realize that due to the inherent complex scale ambiguity in the channel estimate, $\boldsymbol{B}$ is not unique. Because of the condition $\|\boldsymbol{b}_i\|_2 = 1$ in the solution of (9), each column $\boldsymbol{b}_i$ is subject to a multiplicative phase ambiguity of the form $e^{j\gamma_k}$ [see (7)]. Fast variation of this phase ambiguity in frequency can cause the resulting time-domain estimate $\hat{\boldsymbol{H}}(t)$ of the channel to be excessively long. By initializing the algorithm in the manner proposed, this phase ambiguity tends to vary smoothly with frequency, therefore creating an $\hat{\boldsymbol{H}}(t)$ which can be of moderate length. As shown in our simulations, the resulting composite mixing-unmixing system is then more localized in time. A channel with a localized impulse response is known to be less susceptible to degradation in audio quality due to reverberative effects.

It is worthy of note, that in the case where $J > N$, the number of sources N can be estimated from the number of elements of $\mathbf{\Lambda}$ which are significantly different from zero.

V. Resolving Permutations

One potential problem with the cost function in (9) is that it is insensitive to permutations of the columns of $\mathbf{B}(\omega_k)$. More specifically, if $\mathbf{B}_{opt}(\omega_k)$ is an optimum solution to (9) then $\mathbf{B}_{opt}(\omega_k)\mathbf{\Pi}(\omega_k)$, where $\mathbf{\Pi}(\omega_k)$ is an arbitrary permutation matrix for each ω_k, will also be a optimum solution. Since in general $\mathbf{\Pi}(\omega_k)$ can vary with frequency, poor overall separation performance will result.

In this section, we suggest a solution for solving the permutation problem for arbitrary N, that exploits the cross-frequency correlation between diagonal values of $\mathbf{\Lambda}(\omega_k, m)$ and $\mathbf{\Lambda}(\omega_{k+1}, m)$ given in (9). Similar ideas have been reported previously in [3], [11]. Notice that $\mathbf{\Lambda}(\omega_k, m)$ can be considered an estimate of the sources' cross-power spectral density at epoch m. When the sources are speech signals, the temporal trajectories of the power spectral density of speech, known as spectrum modulation of speech, are correlated across the frequency spectrum. This follows because with speech, when the talker is louder, all spectral components of the signal tend to increase in level, and vice-versa. This co-modulation of the frequency components of an audio signal is a common phenomenon, because temporally varying attenuation is often broadband.

Assume that $\mathbf{\Lambda}(\omega_k, m)$, $m = 0, \ldots, M-1$, represents the estimated cross-spectral density of the sources at frequency bin ω_k. We want to adjust the permutation at frequency ω_j such that it has the same permutation as frequency bin ω_k. To do so, we first calculate the cross frequency correlation between the diagonal elements of $\mathbf{\Lambda}(\omega_j, m)$ and $\mathbf{\Lambda}(\omega_k, m)$ using following measure:

$$\rho_{qp}(\omega_k, \omega_j) = \frac{\sum_{m=0}^{M-1} \lambda_q(\omega_k, m)\lambda_p(\omega_j, m)}{\sqrt{\sum_{m=0}^{M-1} \lambda_q^2(\omega_k, m)}\sqrt{\sum_{m=0}^{M-1} \lambda_p^2(\omega_j, m)}} \tag{23}$$

where $\rho_{qp}(\omega_k, \omega_j)$ represents the cross frequency correlation between $\lambda_q(\omega_k, m)$, the q_{th} diagonal element of $\mathbf{\Lambda}(\omega_k, m)$, and $\lambda_p(\omega_j, m)$, the p_{th} diagonal element of $\mathbf{\Lambda}(\omega_j, m)$. To determine whether frequency bins ω_k and ω_j have the same permutation, we perform the following test. Consider the statistic r given by

$$r = \frac{\rho_{qq}(\omega_k, \omega_j) + \rho_{pp}(\omega_k, \omega_j)}{\rho_{pq}(\omega_k, \omega_j) + \rho_{qp}(\omega_k, \omega_j)}. \tag{24}$$

If the permutations between bin ω_k and ω_j are the same, then r should be a large number; conversely, r should be small if the permutations are incorrect. Therefore we change the permutations at one of the frequency bins ω_k and ω_j if $r < 1$. We apply the above test to all frequency bins to detect and correct wrong permutations.

In general, when number of sources is greater than two, the permutation matrix can be estimated by solving the following discrete optimization problem:

$$\max_{\mathbf{\Pi}(\omega_k)\in\mathcal{P}} \text{Trace}\left(\mathbf{\Pi}(\omega_k)\mathbf{E}(\omega_k)\mathbf{E}^T(\omega_j)\right) \tag{25}$$

where $\mathcal{P}$ is the set of $N\times N$ permutation matrices including the identity matrix, and $\mathbf{E}(\omega_k)$ is an $N\times M$ matrix given as

$$\mathbf{E}(\omega_k) = \left(\sum_{m=0}^{M-1}\mathbf{\Lambda}^2(\omega_k,m)\right)^{-\frac{1}{2}} \times \begin{pmatrix} \lambda_1(\omega_k,0) & \dots & \lambda_1(\omega_k,M-1) \\ \vdots & \ddots & \vdots \\ \lambda_N(\omega_k,0) & \dots & \lambda_N(\omega_k,M-1)\end{pmatrix}. \tag{26}$$

The discrete optimization criterion in (25) can be solved by enumerating over all possible selections for $\mathbf{\Pi}(\omega_k)$. This means for a set of $N\times N$ permutation matrices, the criterion in (25) must be calculated $N!$ times to find the optimum solution for $\mathbf{\Pi}(\omega_k)$. For large values of N this may not be computationally efficient. A more computationally efficient but suboptimal approach to estimate the permutation matrix between frequency bins ω_k and ω_j is given by the following algorithm.

Adjusting Permutations

1) Initialize the $N \times N$ matrix $\mathbf{\Pi}(\omega_k)$ to an all zeros matrix.
2) Set up matrices $\mathbf{E}(\omega_j)$ and $\mathbf{E}(\omega_k)$, where $\mathbf{E}(\cdot)$ is given by (26).
3) Form the multiplication $\mathbf{T}_{kj} = \mathbf{E}(\omega_k)\mathbf{E}^T(\omega_j)$
4) Find the row number r_{max} and column number c_{max} corresponding to the element of $\mathbf{T}$ with largest absolute value. Zero all elements of $\mathbf{T}$ corresponding to these row and column numbers and set $\mathbf{\Pi}_k(c_{max},r_{max}) = 1$, where the symbol $\mathbf{\Pi}_k$ is used in place of $\mathbf{\Pi}(\omega_k)$.
5) Recursively repeat the previous step for the remaining elements of matrix $\mathbf{T}_{kj}$ until only one nonzero element remains. Set

$$\mathbf{\Pi}_k(c_f,r_f)=1 \tag{27}$$

where r_f and c_f are the corresponding row and column numbers of the remaining nonzero element.

Notice that the above algorithm calculates the permutation matrix $\mathbf{\Pi}(\omega_k)$ between two frequency bins ω_k and ω_j. To obtain a uniform permutation across the whole frequency spectrum, we need to apply the above algorithm repeatedly to all pairs of frequency bins. One way of doing this is to adjust the permutation between adjacent frequency bins in a sequential order where, for example, starting from frequency bin ω_0 we adjust the permutation of each bin relative to its previous bin. This approach, although simple, has a major drawback as explained as follows. Consider the situation where an error is made in estimating the correct permutation matrix for frequency bin ω_k. In this case, all frequency bins placed after ω_k will receive a different permutation than the ones placed before ω_k. In the worst case scenario we will have half of the frequency bins with one permutation and the other half with a different permutation, which will result in very poor or no separation of the output signal. To prevent such a catastrophic situation we propose following dyadic hierarchical sorting scheme to sort the permutations across all frequency bins.

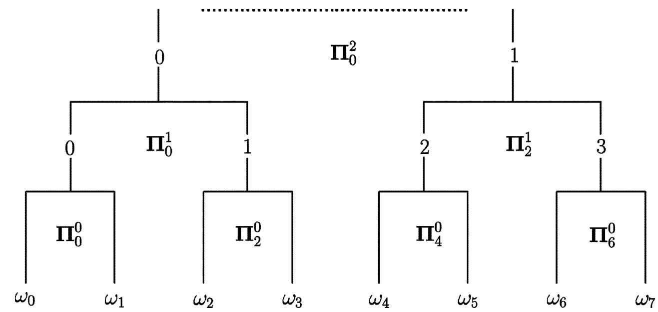

Fig. 1. Example of the dyadic permutation sorting algorithm for the case when the total number of frequency bins K =8.

Dyadic Sorting Algorithm

Initialization: With reference to Fig. 1, let l be the hierarchical level index,[6] $l=0,\dots,\log_2(K)-1$, where the zeroth level is at the bottom of Fig. 1. Initialize $l=0$. Define $\mathbf{\Sigma}_k^0(m)=\mathbf{\Lambda}(\omega_k,m)$, $k=0,\dots,K-1$.

FOR $l=0,\dots,\log_2(K)-1$, BEGIN

1) Divide the available frequency bins at hierarchical level l into groups of two bins, with group index p, $p=0,1,\dots,K/(2^{l+1})-1$.
2) Let $\mathbf{\Pi}^l(\omega_j)$ for each $j \in [2p,2p+1]$ be the permutation matrix for the hierarchical level l and group p estimated from the proposed permutation algorithm. Then for all $p=0,\dots,K/(2^l)$ and $j\in[2p,2p+1]$, update the order of diagonal values of $\mathbf{\Sigma}_j^0(m)$ using

$$\mathbf{\Sigma}_j^l(m) := \mathbf{\Pi}^l(\omega_j)\mathbf{\Sigma}_j^l(m)\mathbf{\Pi}^{lT}(\omega_j)$$

3) Update the order of the columns of $\mathbf{B}(\omega_j)$ using

$$\mathbf{B}(\omega_j) := \mathbf{B}(\omega_j)\mathbf{\Pi}^l(\omega_j)$$

4) For each group calculate

$$\mathbf{\Sigma}_p^{l+1}(m) := \mathbf{\Sigma}_{2p}^l(m)+\mathbf{\Sigma}_{2p+1}^l(m), \quad l\neq\log_2(K)-1 \tag{28}$$

END

[6]Here we assume that K, the total number of the frequency bins, is an integer power of 2.

The success of the proposed dyadic sorting algorithm can be explained for the $N = 2$ case as follows. As the process moves up the hierarchy, the statistics comprising the numerator and denominator of (24) accumulate more data samples, as is made evident by (28). Thus, the probability distribution governing these statistics decreases in variance, with the result that the probability of error in (24) diminishes as the hierarchy is ascended. This argument is easily extended to the $N > 2$ case. Thus, we can say that the dominant error mechanism is at the lower levels of the hierarchy, where the impact of a permutation error is not catastrophic.

VI. Simulation Results

In this section, we present two examples using a synthetic FIR mixing system and synthetic sources, to demonstrate the performance of the proposed permutation algorithm, and a 3 × 3 mixing system consisting of IIR filters.

A. Example I: FIR Convolutive Mixing

The objective of this first simulation is to characterize the performance of the algorithm under a controlled mixing environment. For this purpose we use an 8-tap FIR convolutive mixing system whose impulse responses are selected randomly from a uniform distribution with zero mean and unit variance. For the sources, we use two independent white Gaussian processes, multiplied by slowly varying sine and cosine signals respectively to create the required nonstationary effect. In this example, the epoch size was kept at 500 and the total data length was varied between 10 000 and 50 000 samples, corresponding to M, the number of epochs, ranging between 20 to 100 epochs. White Gaussian noise was added to the output of the system at a level corresponding to the desired value of averaged SNR over all epochs.[7] At each epoch, 128-point FFTs, applied to time segments overlapping by 50%, weighted by Hanning windows were used to estimate the cross-spectral density matrices. For the joint diagonalization algorithm, for all frequency bins for this specific case only, we chose the initial estimate of the channel to be an identity matrix.[8]

Let $\mathbf{C}(\omega_k)$ represent the composite mixing-unmixing system frequency response

$$\mathbf{C}(\omega_k) = \mathbf{W}(\omega_k)\mathbf{H}(\omega_k) \tag{29}$$

whose ij_{th} element is $c_{ij}(\omega_k)$. To measure the separation performance, we evaluate the *signal to interference ratio* (SIR), given by the following formula:

$$\mathrm{SIR}(i) = \frac{\sum_{q=0}^{M_c-1} \max_j \left(S_{ij}^q\right)}{\sum_{q=0}^{M_c-1} \left[\sum_{j=1}^{N} S_{ij}^q\right] - \max_j \left(S_{ij}^q\right)} \tag{30}$$

where $S_{ij}^q = \sum_{k=0}^{K-1} |c_{ij}^q(\omega_k)|^2$, q is an index to the q_{th} Monte Carlo run and the quantity M_c is the total number of Monte Carlo runs. Notice that (30) implicitly measures the cross channel interference for each output of the separating system. Here, at each output, the desired signal is the separated source that has the maximum power and the interference is the contribution from the other sources. Fig. 2 shows the variation of each output's SIR with M, the number of epochs for a fixed signal-to-noise ratio (SNR $= 20$ dB). As can be seen from this figure, by increasing the number of epochs, which corresponds to increasing the data length, the output SIR improves (increases). Also, Fig. 3 shows how the separation performance changes with observed signal's signal-to-noise ratio for a fixed number of epochs, M =50.

We now include a second example, which shows how the algorithm can perform when there are more than two sources, and when the mixing system is a stable matrix of infinite impulse response (IIR) filters. Most existing algorithms consider only the case where the mixing system consists only of finite impulse response (FIR) filters. This simulation shows that the proposed algorithm, given enough data samples at each epoch, can perform well for the IIR case. For this example, we use a 3 × 3 IIR

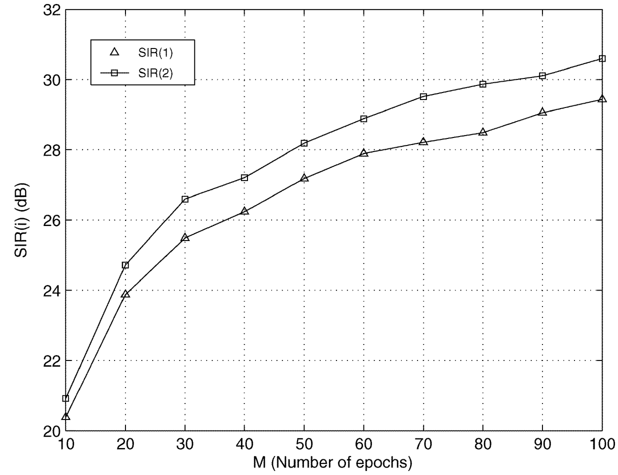

Fig. 2. Example I: SIR versus M, the number of epochs, for SNR $= 20$ dB, using $M_c = 50$ Monte Carlo runs.

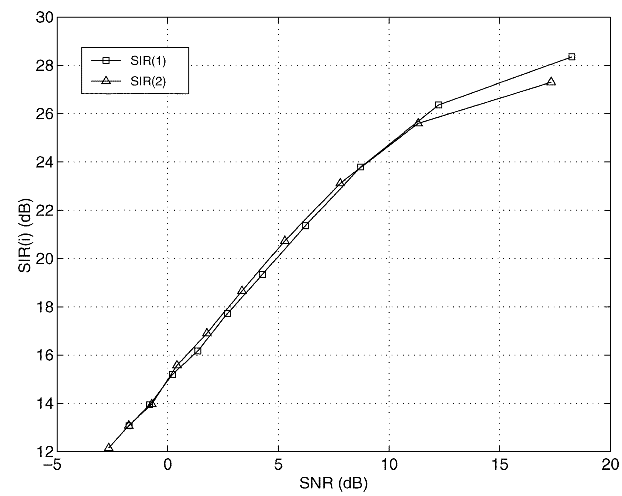

Fig. 3. Example I: SIR versus SNR, for m $= 50$, using $M_c = 50$ Monte Carlo runs.

[7]The power of the noise was kept constant at all epochs.

[8]The initialization method described in the previous section can of course be used in this example. Nevertheless the motivation for using an identity matrix for initialization in this specific example is to demonstrate the desirable convergence properties of the algorithm, even if we do not know a good initialization point.

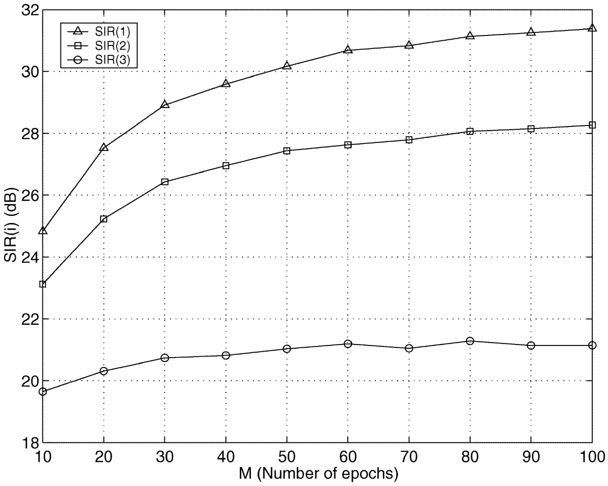

Fig. 4. Example 2: SIR versus M, for SNR $= 20$ dB and using $M_c = 50$ Monte Carlo runs.

mixing system where the impulse response of the $ij_{\rm th}$ element is given as

$$h_{ij} = \frac{b_{ij}}{1 - \alpha_{ij} z^{-1}} \tag{31}$$

where $0 < \alpha_{ij} < 1$ and b_{ij} are given as

$$\alpha_{ij} \in \begin{bmatrix} 0.555 & 0.761 & 0.198 \\ 0.921 & 0.432 & 0.633 \\ 0.144 & 0.188 & 0.231 \end{bmatrix}$$
$$b_{ij} \in \begin{bmatrix} -1.50 & -0.19 & 1.21 \\ -0.52 & 2.53 & -0.27 \\ -0.95 & -0.03 & 0.38 \end{bmatrix}. \tag{32}$$

We use three sources, where the first two are the same as in example I and the third one is a white Gaussian signal with a slowly decaying exponential envelope. All the sources are statistically independent. The sources were mixed using the Matlab "filter" command, and similar to example I, white Gaussian noise was added to the mixture at a variable power level corresponding to the desired signal-to-noise ratio.

For this example, the epoch size was increased to 4000 data samples, and at each epoch 256-point FFTs were used, applied to time segments overlapping by 80%, weighted by Hanning windows, to estimate the cross-spectral density matrices. For each output, the signal to interference ratio was measured using the criterion of (30). The results are given in Fig. 4, which shows the SIR versus M, for a 20-dB signal-to-noise ratio.

VII. Real Room Experiments

In this section, we present the results of the proposed algorithm in a real reverberant environment. All recordings were done using an 8.0-kHz, 16-bit sampling format.

A. Real Room Experiment I

The first set of experiments was conducted in an office room using $N = 2$ loudspeakers for the sources and $J = 4$ omnidirectional microphones for the sensors. The configuration of the experiment, showing all distances etc., is shown in Fig. 5. The two sources were created by catenating independent multiple

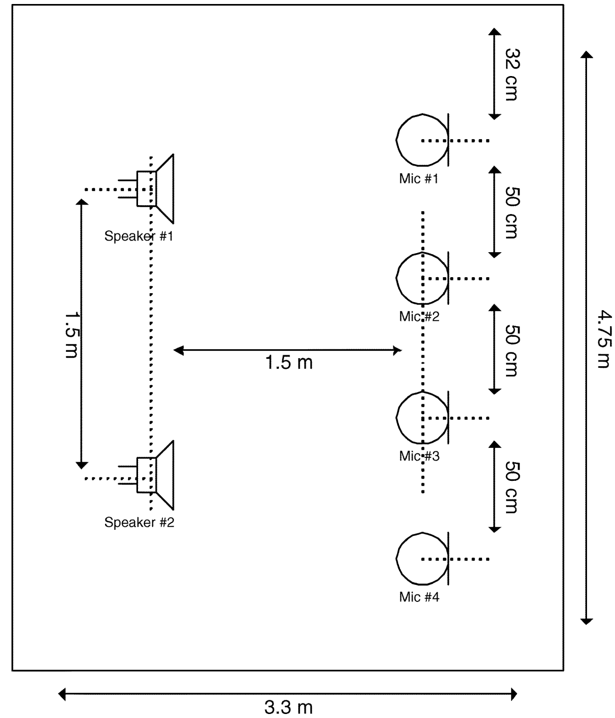

Fig. 5. Recording setup for the office room. (Experiment I.).

speech segments from the TIMIT speech database. The speech signals then were played simultaneously through the two loudspeakers with approximately the same sound volume.

To quantify the separation performance, we measure the SIR at each microphone (i.e., the input to the algorithm) and at each output of the algorithm. In this case however, we cannot use (30) directly, since it requires knowledge of the channel, which is unknown in this case. Therefore to estimate the SIR, using the same experimental setup, white noise signals were played through each loudspeaker one at a time (i.e., only one source was active at each time). Let $\hat{\sigma}_x^2(x_i, s_j) = \sum_{t=0}^{T-1} x_i^2(t)$, where T is the number of data samples, represent the power of the recorded signal at the i_{th} microphone when only speaker j is active and all other speakers (sources) are inactive. The signal-to-interference ratio of the recorded signal $x_i(t)$ at the i_{th} microphone can be estimated using

$$\mathrm{SIR}_x(i) = \frac{\max_j \hat{\sigma}_x^2(x_i, s_j)}{\sum_{j=1}^{N} \hat{\sigma}_x^2(x_i, s_j) - \max_j \hat{\sigma}_x^2(x_i, s_j)}. \tag{33}$$

Using above formula, we can also measure $\mathrm{SIR}_y(i)$, the SIR of the ith output of the separating network, by substituting $\hat{\sigma}_x^2(x_i, s_j)$ with $\hat{\sigma}_y^2(y_i, s_j)$, the power of the signal at the i_{th} output when only source j is active.

To perform the separation, in a manner similar to example I, we first divided the recorded signal into multiple time segments (epochs), where we chose the size of each epoch to be 10 000 data samples long. For each epoch, the cross spectral density matrices were calculated, with the number of FFT-points K equal to 4096, and with the overlap-percentage equal to 80%. Fig. 6 shows the output $\mathrm{SIR}_y(i)$ versus M, the number of epochs, for $i = 1, 2$. As can be seen for $M = 100$, an average SIR of more than 20 dB is reached for each output.

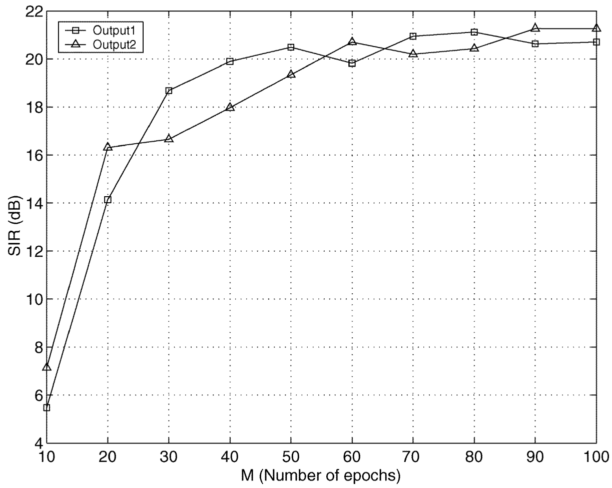

Fig. 6. Results of separation performance for recordings in an office room: $\mathrm{SIR}_y i$ versus M, the number of epochs, for K =4096, i =1, 2.

TABLE I
INPUT SIRs FOR THE RECORDED SIGNALS IN AN OFFICE ENVIRONMENT

	MIC 1	MIC 2	MIC 3	MIC 4
SIR_x	2.6945 dB	1.2282 dB	0.3266 dB	1.4031 dB

As a reference, Table I shows the input SIRs corresponding to the recorded signals. By comparing the SIRs before and after applying the separating algorithm, it can be seen that the output SIRs have been improved by roughly 19 to 20 dB. In this experiment, on average, at each frequency bin the ALS algorithm converged within 30 to 40 iterations.

As discussed in Section II, we can recover the sources only up to a frequency dependent scaling ambiguity using the proposed joint diagonalization algorithm. The effect of this unknown scaling ambiguity can significantly deteriorate the quality of the separated audio signals. However, our listening tests reveal that the sequential initialization procedure discussed in Section IV-A results in improved quality of the separated speech signals in these experiments.[9] To demonstrate the effect of the initialization procedure on the impulse response of the separating network, we calculate the impulse response of the separating matrix with and without using the proposed initialization method[10] Fig. 7 shows the results for one of the impulse responses of the separating network. As can be seen from the figure, the tail of the impulse response of the separating matrix obtained using the new initialization procedure is more suppressed (except for low-frequency artifacts which can easily be removed by high-pass filtering) compared to the tail of the impulse response of the separating matrix calculated without using the sequential initialization procedure. It is well known that the tails do contribute to the distortion of the separated audio signal and the more suppressed they are, the better is the perceptual quality of the separated signals.

[9]To hear the recordings and also the separation results, please refer to the following website: http://ieeexplore.ieee.org.

[10]When the proposed initialization procedure is not used, the ALS procedure is initialized to an identity matrix at each frequency.

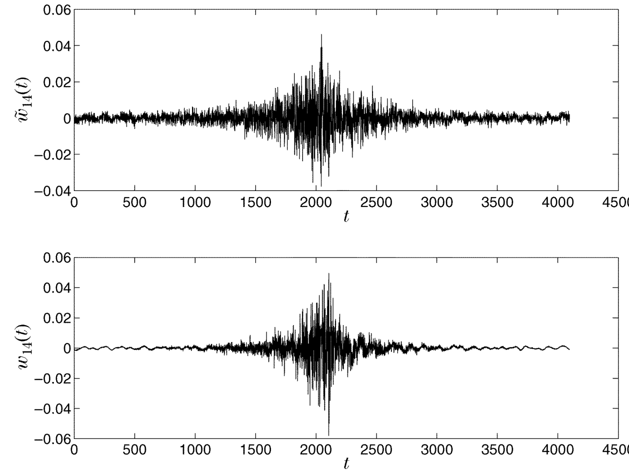

Fig. 7. Impulse response of the separating matrix without $[\tilde{w}_{14}(t)]$ and with $(w_{14}(t))$ the sequential initialization procedure being applied.

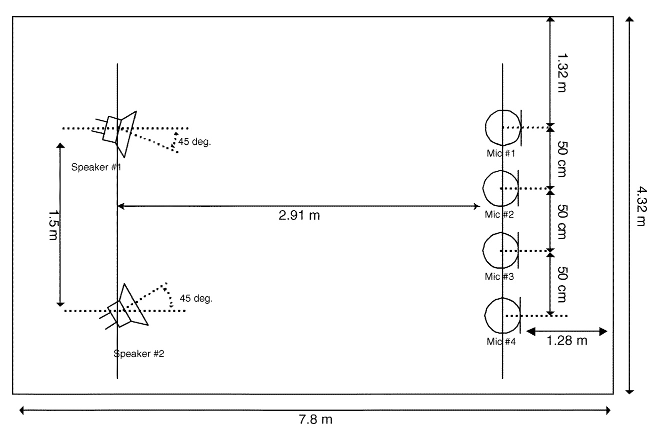

Fig. 8. Real room experiment II: recording setup in a conference room.

B. Real Room Experiment II

In this next set of experiments, we perform real recordings in a highly reverberant conference room. The recording setup is similar to the previous experiment, except that the room dimensions and the distance between the microphones and speakers are increased for this experiment (Fig. 8). The reverberation characteristics of the two rooms used in these experiments are shown in Fig. 9, which shows the reverberation time[11] of the room versus frequency. As can be seen from the figure, on average the reverberation time of the conference room, used in this experiment, is much higher than the reverberation time of the office room, used in the previous experiment. Due to the long reverberation time of the conference room, we expect that in this case more frequency bins are needed to estimate the cross spectral density matrices of the recorded signals. In this experiment, the epoch size was kept the same as in the previous experiment (10,000 samples). Fig. 10 shows how the performance of the algorithm improves by increasing the number of frequency bins used to estimate the cross spectral density matrices. (Zero-padding was used if K was greater than the number of samples in an epoch (10,000)). As can be seen, a total number

[11]The reverberation time in a room at a given frequency is the time required for the mean-square sound pressure in that room to decay from a steady state value by 60 dB after the sound suddenly ceases(see also [34]).

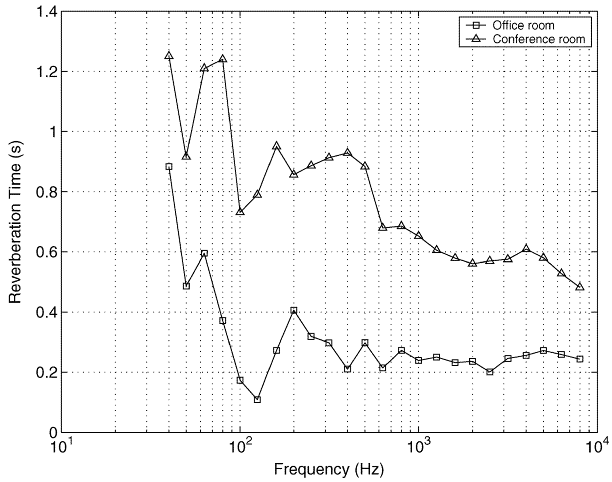

Fig. 9. Reverberation times of the rooms used in experiments I and II.

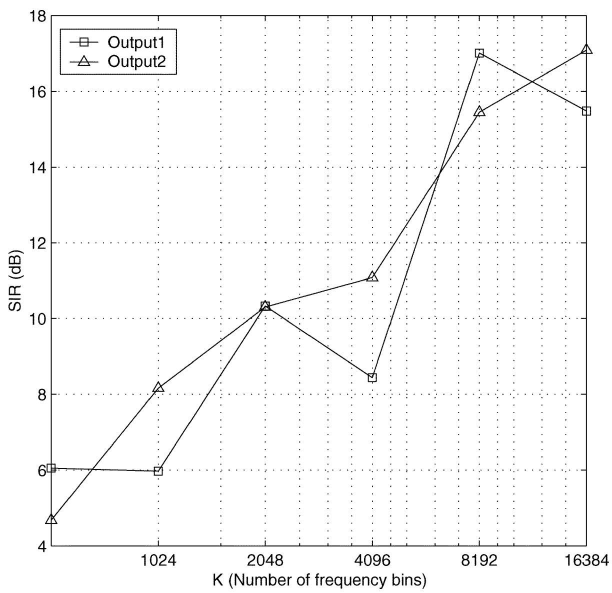

Fig. 10. Separation performance versus number of frequency bins (K) for recordings in a conference room, $M = 140$.

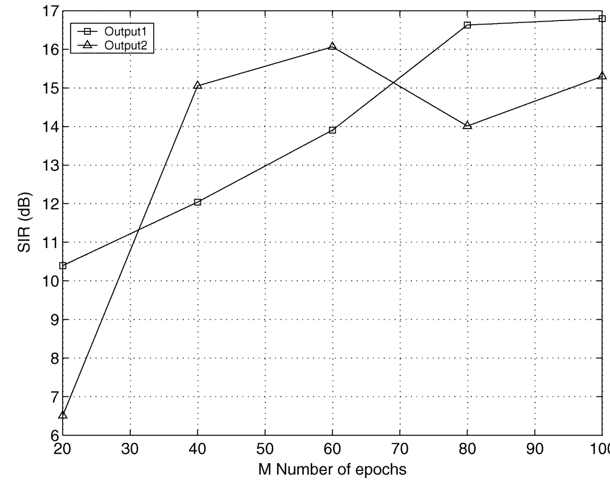

Fig. 11. Separation results for recordings in a conference room: SIR versus M for $K = 16\,384$.

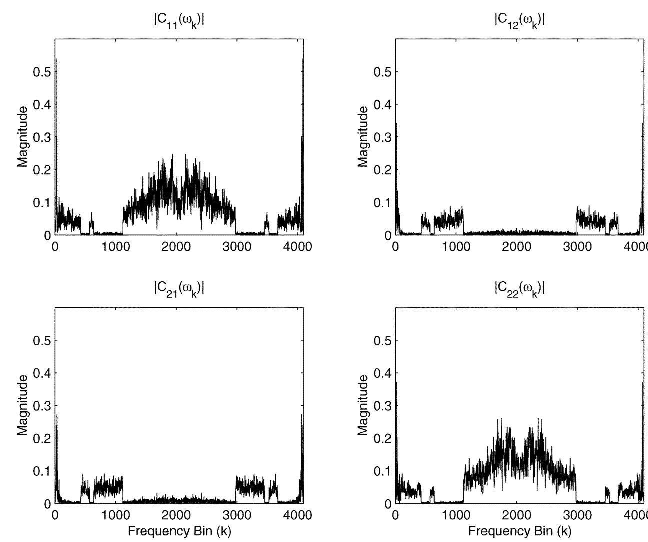

Fig. 12. Each subfigure gives the magnitude response of a c_{ij} as defined in (29), for $i, j \in [1, 2]$ before applying the permutation algorithm.

of 8192 frequency bins are needed to achieve a separation performance between 14–16 dB. The system performance versus M is shown in Fig. 11. Also in this case, convergence of the ALS algorithm was obtained within 30–40 iterations.

It is to be noted that because these experiments were conducted using loudspeakers as sources, the room response was guaranteed stationary over the entire time required to observe the data. In a real environment, if the sources are speech, it would be expected that there would be some movement of the source positions during the data observation interval. This would result in a degradation in performance.

C. Performance of the Permutation Algorithm

In this section, we investigate the behavior of the proposed permutation algorithm in real room environments. Measured room impulse responses corresponding to a variety of microphone/loudspeaker placements in a 3.5 m $\times$ 7 m $\times$ 3 m conference room available from [35] were downloaded. The impulse responses are on the order of 250–300 ms in duration and were resampled to an 8-kHz sampling rate. These impulse responses were then used to construct a 2×2 mixing system. The same two speech sequences used for the previous real-room experiments were fed through this mixing system to produce the mixed signals used for this experiment.

The mixtures were then separated using the proposed algorithm, using the values $K = 4096$ and $M = 100$, and a data size of 10 000 samples for each epoch. Separation results before and after applying the permutation algorithm are shown in Figs. 12 and 13, respectively. Each subfigure gives the magnitude response of a c_{ij}, as defined in (29), for $i, j \in [1, 2]$. The presence of a discontinuity at a specific frequency in all four subfigures of Fig. 12 indicates a permutation mis-adjustment in the magnitude response of $\boldsymbol{C}$. As can be seen, there are several instances where permutation errors exist. However, as shown in Fig. 13, there are no visible discontinuities after application of the permutation algorithm, indicating the permutations have been correctly resolved.

To further illustrate the performance, we conducted three additional experiments, one of which used the same mixing

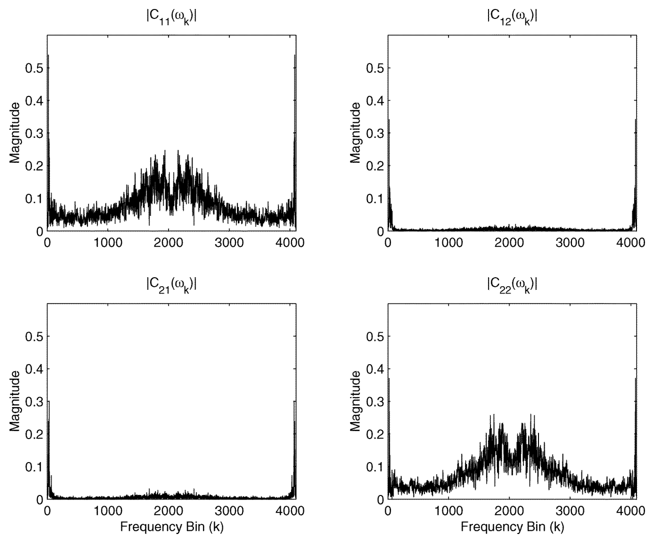

Fig. 13. Same as Fig. 12, except after applying the permutation algorithm.

TABLE II
OUTPUT SIR BEFORE AND AFTER APPLYING THE PERMUTATION ALGORITHM USING ACOUSTIC IMPULSE RESPONSES MEASURED IN A REAL ROOM

Experiment	1		2		3	
Output SIR (dB)	SIR(1)	SIR(2)	SIR(1)	SIR(2)	SIR(1)	SIR(2)
Before applying the permutation algorithm	9.4	8.6	10.4	9.5	6.1	6.3
After applying the permutation algorithm	16.2	15.7	17.5	16.8	15.9	17.9

system as in the previous example, and two which corresponded to additional, similar mixing systems from the same website. Mixed signals were then generated as in the previous experiment. These mixed signals were then separated using the proposed separation algorithm, using the same parameter values as in the previous case. Here, the signal-to-interference ratio given by (30) was used to evaluate the separation performance before and after application of the permutation algorithm. The results are shown in Table II. As can be seen, the improvements in separation performance due to the permutation algorithm are between approximately 7–11 dB. These results indicate the effectiveness of the proposed permutation algorithm using mixing systems constructed from measurements taken in real rooms.

D. Comparisons With Existing Methods

In this section we compare the performance of the proposed method versus that of Parra's method [4]. The proposed algorithm was evaluated using the data[12] available from the authors of [36], and compared to the results given by the lower curve in Fig. 1 of [4]. These latter results are obtained using the same data set. The measurements consist of 10 s of simultaneous speech, and were recorded at 16 Khz in a $4.2\ \text{m} \times 5.5\ \text{m} \times 3.1\ \text{m}$ room, using omni-directional microphones.

For the proposed algorithm, the epoch size was kept at 10 000 samples. The performance of the proposed method versus K, using the performance criterion of [4], is shown in Table III. In contrast, Parra's method yields at best approximately 4.8 dB of separation for the same conditions. It is apparent therefore

[12]This data is available from http://www2.ele.tue.nl/ica99.realworld.html.

TABLE III
SEPARATION PERFORMANCE OF THE PROPOSED ALGORITHM FOR $M = 16$, USING THE DATA AVAILABLE FROM THE AUTHORS OF [36]. THE DATA ARE EXPRESSED IN DECIBELS

K	Output 1	Output 2
512	0.17	1.86
1024	1.48	3.78
2048	4.1644	2.1219
4096	9.7591	9.3944

that the proposed method compares favorably with the former method, at least under the present test environment and for suitably large K. However, it is to be noted that Parra's method behaves significantly better than 4.8 dB when unidirectional microphones are used. Their results show that up to 16–17 dB of separation can be obtained with real-room recordings in a $3\ \text{m} \times 3.6\ \text{m} \times 2.3\ \text{m}$ room at 8 kHz, with 30 s of speech.

Reasons for the improved performance of the proposed method over Parra's approach in the scenario described above include the following. Firstly, Parra's method constrains the length of the FIR demixing system to resolve permutation errors. This approach has been shown to be ineffective for resolving permutations, as independently verified by [25]. When the demixing system length for the method of [4] is greater than the optimum, its performance is dominated by permutation error, and when it is less, the demixing system does not have sufficient degrees of freedom to adequately separate the sources. Second, the proposed least-squares based ALS algorithm has improved performance over the gradient-based method used by [4].

VIII. CONCLUSION

In this paper, we discussed a new recursive algorithm for BSS of convolved nonstationary sources. We used the set of the observed signals' cross-spectral density matrices evaluated over different time segments to recover the sources up to a frequency dependent scaling and permutation ambiguity, using a joint diagonalization procedure. A two-stage frequency-domain algorithm to estimate the un-mixing filters was proposed. A novel procedure for resolving the frequency-dependent permutation ambiguities, and an initialization method which significantly improves the perceptual quality of the separated audio signals was also proposed.

The performance of the new algorithm using speech sources with real mixing systems under different mixing scenarios was demonstrated. It was shown that the algorithm performs well in real environments with at least 15-dB separation in a reverberant conference room scenario and at least 20 dB for a smaller office room.

A significant feature of the proposed method is that unlike most existing algorithms, no assumptions on the structure of the mixing system were made; i.e., the elements of the mixing system can be FIR or IIR filters. Another important feature of the algorithm is use of a recursive least-squares algorithm which has the advantage of fast convergence with no required parameter tuning.

APPENDIX
COMPUTATION OF THE CROSS-SPECTRAL DENSITY MATRICES

Given the sequence $\boldsymbol{x}(t)$ defined by (1), the cross-spectral density matrix $\tilde{\boldsymbol{P}}_x(\omega)$ is defined as

$$\tilde{\boldsymbol{P}}_x(\omega) = \int_{-\infty}^{\infty} \mathbf{R}_x(\tau) \exp(-j\omega\tau) d\tau \tag{34}$$

where $\mathbf{R}_x(\tau)$ is the spatial covariance matrix, defined as

$$\mathbf{R}_x(\tau) = \mathrm{E}\left(\boldsymbol{x}(t+\tau)\boldsymbol{x}^T(t)\right) \tag{35}$$

where $\mathrm{E}(\cdot)$ is the expectation operator. To estimate the observed signals' cross-spectral density matrices, $\tilde{\boldsymbol{P}}_x(\omega_k, m)$, $m = 0, \ldots, M-1$, $k = 0, \ldots, K-1$, we use the Welch periodogram method [37] to approximate (34). We divide the observed sequence into M epochs, where stationarity can be assumed within the epoch but not over more than one epoch. We then apply the following formula to estimate the cross-spectral density matrix at the m_{th} epoch

$$\hat{\mathbf{P}}(\omega_k, m) = \frac{1}{N_s} \sum_{i=0}^{N_s-1} \mathbf{x}_i(\omega_k, m)\mathbf{x}_i^{\dagger}(\omega_k, m) \tag{36}$$

where

$$\mathbf{x}_i(\omega_k, m) = \sum_{n=-\infty}^{\infty} \mathbf{x}(n) w(n - iT_s - mT_b) e^{-j\omega_k n} \qquad k = 0, \ldots, K-1 \tag{37}$$

where N_s is the number of overlapping windows inside each epoch, T_b is the size of each epoch, T_s is the time shift between two overlapping windows, K is the number of frequency bins, and $w(n)$ is the windowing sequence. Note that $\mathbf{x}_i(\omega_k, m)$ in (37) is computed using the FFT. A de-noised estimate can be obtained by forming $\hat{\mathbf{P}}(\omega_k, m) - \lambda_{min}\boldsymbol{I}$, where λ_{min} is the minimum eigenvalue. Due to the finite data approximation, and the smoothing effect of the Welch periodogram, the resulting $\hat{\mathbf{P}}(\omega_k, m)$ only satisfies (4) approximately. The approximation is valid when the window length is much larger than L.

For convenience, the estimated cross spectral density matrices can be normalized, and so for our computations we set

$$\hat{\mathbf{P}}_x(\omega_k, m) = \frac{\hat{\mathbf{P}}(\omega_k, m)}{\left\|\hat{\mathbf{P}}(\omega_k, m)\right\|_F}. \tag{38}$$

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